Manufacturing Processes
Unit - 1

Manufacturing
→ process of converting raw material into desired finished products in economical cost.
→ incompasses design & fabrication of goods with the help of various production method & techniques.

Input → Transform → Output

- Input:
  - Raw material
  - Capital
  - Machines
  - Technology
- Output: Desired Finished Products

→ Economical & Technological Consideration
Economical → cost, availability, transport
Technological → automating machines,

Property of Materials
- Ductile (ability of material to be drawn in wire)
- Malleable (ability of material to convert in thin sheets)
- Strength
- Tensile ←□→
- Compress →□←
- Stiffness

Classification of Manufacturing Process
① Casting
- produce metal/alloy components of desired shape by pouring molten metal into cavity.

② Forging (Forming)
- modify the shape & size of the product/material.

③ Fabrication (Joining Process)
- joining work pieces
- Types → Strength: Ⓐ → Ⓒ
  Ⓐ Welding Ⓑ Brazing Ⓒ Soldering

④ Machining (To remove excess portion)
- removal of certain areas from the part to attain required shape.
- Chips → formed by "interact" of cutting tools with material using machines.

Casting
- process of producing metal/alloy components of desired shape by pouring molten metal into mould cavity & allowing it to cool & solidify.
- solidified piece of metal → Casted Product

Pattern
Replica of an object → mould cavity

Material used for making pattern
① Wood
Advantages:
- Easy working
- Available
- Easy to Handle
- Cheap
- No rusting
- Lighter than metal

Wood: Soft (Pine) — Hard (Teak, Mango, Sisham)

Disadvantages:
- Easily affected by moisture.
- Wear out easily

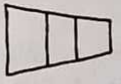
② Metal
Advantages:
- Durable & accurate
- Resistant to abrasion & swelling
- Smooth surface
- Withstand rough handling

Disadvantages:
- Rusting
- Heavy
- Expensive

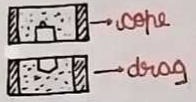
③ Plastic (PVC, Epoxy, Resin)
Advantages:
- Light weight
- Cheap & durable
- No moisture
- Corrosion resistant
- Easy working

Disadvantages:
Non-environmental friendly

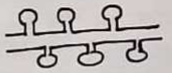
Types of Patterns
① Solid or Single Piece
- Simplest
- one piece + no partition



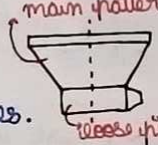
② Split Pattern
- two pieces
- Lower → drag, Upper → cope



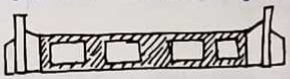
③ Gated Pattern
- produce several small casting at a time



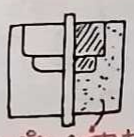
④ Loose Piece Pattern
- projections are made in loose pieces & attached to main pattern by means of pins.



⑤ Skeleton Pattern
- used in large casting to provide outline and size of desired cavity.



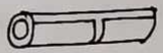
⑦ Sweep Pattern
- used for symmetrical cross section to save time, labour & material



⑧ Follow Board Pattern
- used for fragile & structurally weak pattern to support it & avoid breaking or collapsing under the force of ramming



⑨ Shell Pattern
Hollow pattern like water pipe



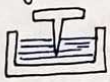
Pattern Allowances

① Shrinkage or contract^{n} allowances
- compensate metal shrinkage that took place when metal solidifies.

Steel → 20 mm/m Cu → 16 mm/m

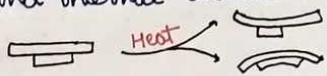
② Machining
- to remove extra material to obtain exact dimension & surface finish

③ Draft or Taper
- provided on vertical surface to facilitate easy removal from mould.
- amount varies from 0.5° to 1.5°



④ Shaking / Ramping
- compensate for enlargement of mould cavity due to excess ramming.
- small, medium casting → neglected
  large casting → cavity size ↓ (little)

⑤ Distortion or camber allowance
- compensate possible distortion of casting due to unequal cooling rates of different sections of casting and internal stress.



Moulding Sand

→ Constituent of Moulding Sand
- Silica (SiO_2) → 86 to 90 %
- Alumina (Al_2O_3) → 4 to 8%
- Iron oxide (Fe_2O_3) → 2 to 5%
- Clay as binder
- Moisture to prevent elasticity

→ Types of Moulding Sand

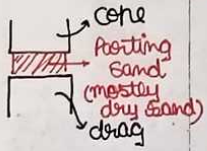
① Green Sand
- mixture of silica sand with 20-30% of clay having total water content from 6 to 10%
- used for small & medium size casting of ferrous & non-ferrous material.

② Dry Sand moisture < 3-4%
- excess moisture is removed by drying or baking.
- used for large casting

③ Facing Sand
- high strength & refractoriness
- consist silica sand without addⁿ of used sand
- used next to pattern surface which come in contact with molten metal.

④ Baking Sand
- used sand & form back of the facing layer,
- due to contact with molten metal & addition of coal dust, its colour is black.

⑤ Parting Sand
- to avoid sticking of sand of one flask to other or to the pattern.



⑥ Loam Sand
- contains clay upto 50%
- for more binding

→ Properties of Moulding Sand

① Refractoriness
→ ability of moulding sand to withstand high temperature without fusing.

② Flowability
- ability to flow over & around the pattern & all portion of moulding flask during ramming.
- increases by adding clay & water

③ Permeability
- ability of sand to allow gases to escape through the sand mould.

④ Cohesiveness
- ability of sand particles to stick together & retain a given mould shape.

⑤ Adhesiveness
- property of sand due to which sand particles are capable of adhering to another body.

⑥ Collapsibility
- ability of sand to collapse mould after casting solidification.

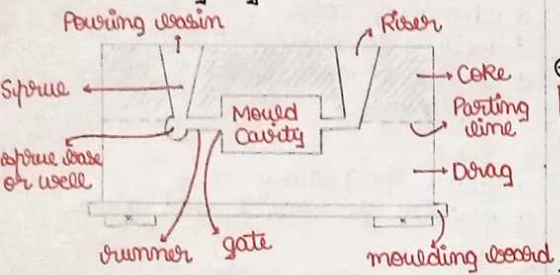
⑦ Chemical Resistivity
- ability of sand doesn't react chemically with molten metal.

⑧ (i) Green Strength
→ strength of sand in moist state

(ii) Dry Strength
→ prevent wash erosion
→ handle pressure during casting.

(iii) Hot Strength
→ ability of sand to withstand high temperature of molten metal during casting.

Gating System



① Pouring basin
- a reservior at the top of the sprue through which molten metal is fed smoothly in sprue.
- maintains the required rate of metal flow and reduces turbulance at sprue entry.

② Sprue
- tapered channel whose bigger end connected to pouring basin and end is connected to sprue base or well.

③ Runner
- passage through which molten metal is carried from sprue base to gate.

④ Gate
- opening through which molten metal is carried into mould cavity.

⑤ Sprue base or well
- one end → sprue & other → runner
- provide uniform flow of molten metal to mould cavity through runner.

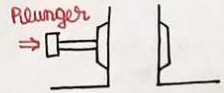
⑥ Riser
- reservior of molten metal through material flows back into mould cavity to compensate any shrinkage during solidification.

Note → Mould → ① Temporary (eg → Casting) ② Permanent (eg → Die Casting)

Steps of Casting

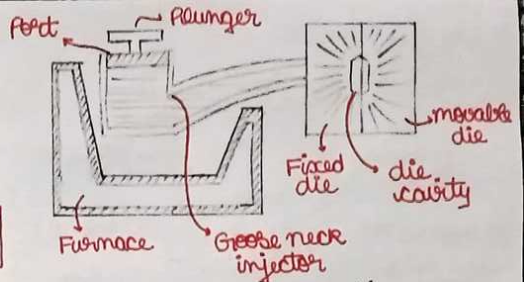
Make Pattern (⊙ Wood ⊙ Metal ⊙ Plastic) → Prepare Moulding Sand → Mould Cavity → Pour molten metal → Solidify → Finishing → Product Ready!

Die Casting
- molten metal is forced with mould cavity under pressure.
- high pressure is due high velocity of injected molten metal.
- help in production of intricate (complex) components at lower cost.

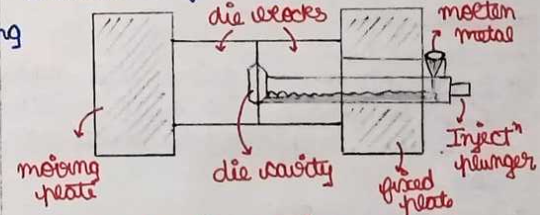


- Two Types →

Ⓐ Hot Chamber die casting
- furnace is integral part of casting machine, to melt the metal in moulding pot fitted in furnace.
- Goose neck → → supply molten metal in die cavity
- During the downward motion of plunger, the port gets closed & plunger applies the pressure on molten metal to flow in die cavity
- After certain time, piston moves upward due to which unused molten metal returns into goose neck
- suitable for zinc, tin, lead & other low melting point alloys.

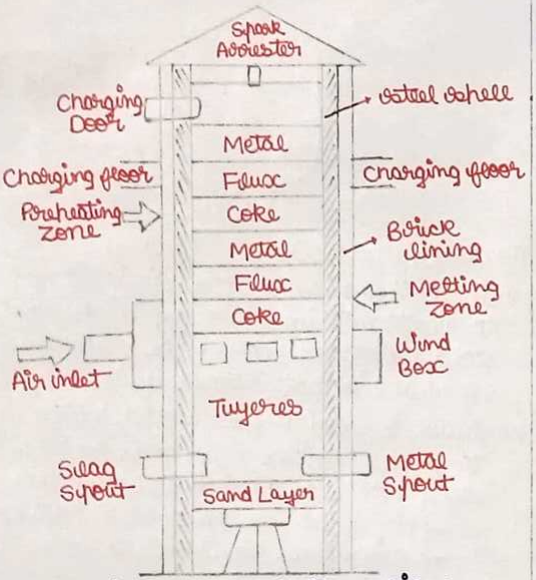


Ⓑ Cold Chamber Die Casting
- In hot chamber, it is difficult to cast materials having high MP, cold chamber solves the problem.
- furnace is not integral part
- ladle is used to fill molten metal to flow into the chamber, plunger reciprocate in horizontal cylinder & apply pressure on molten metal to flow into die cavity to form the casting.
- suitable for high MP alloys.



→ Merit of die casting
- High productive rate
- closed dimensional tolerance can be achieved
- Die has long life
- very good surface finish can be obtained
- Economical for large production
- Thin sections can be cast

Cupola Furnace



- extensively used in its various forms for melting cast iron.
- Initial cost is low.

→ Construction :

- It consist of cylindrical steel shell, lined internally with heat resisting firebricks, supported on cast iron legs.
- Legs → support
- Slag Spout → remove slag from melting iron.
- Tuyeres → enter gas to proper burn of fuel.
- Sand bed → taper form, melted iron comes out easily.
- Spark Arrester → prevent emission from the fireplace.
- Flux → limestone used to remove impurities
- Dropdoor → to clean cupola base

→ Working :

- Heat is generated from burning coke
- When T > MP, metal melts
- Charge consist of pig iron, flux, coke, limestone.
- The working is over the sand
- Flux help to get impurities out, with the help of slag spout.

→ Zones :

Ⓐ Combustion Zone
- Super heating zone
- above tuyeres (150 - 300 mm)
- produces heat for all zones
- T : 1500°C — 1800°C

$$Si + O_2 \longrightarrow SiO_2 + Heat$$
$$C + O_2 \longrightarrow CO_2 + Heat$$

Ⓑ Reducing Zone
- between combustion zone and top of coke bed
- T : ↓ 1200°C

$$CO_2 + C \longrightarrow 2CO + Heat$$

Ⓒ Melting Zone
- above reducing till first metal layer
- metal layer → molten state → collected into well
- T : 1600°C

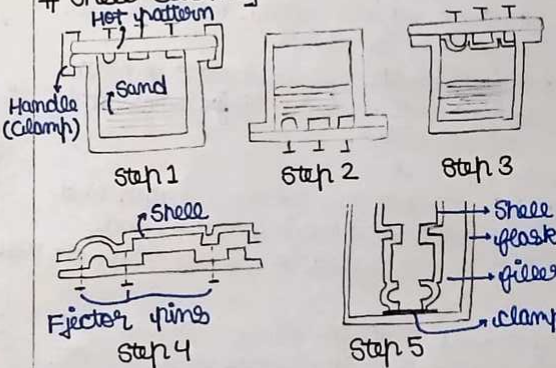
$$2Fe + 2CO \longrightarrow Fe_3C + CO_2$$

Ⓓ Preheating Zone
- charging zone
- melting to charging door
- layers of coke, flux & metal
- T : 1090°C

Ⓔ Stack
- above preheating zone
- provide passage to hot gases

Ⓕ Well
- collect molten metal

Shell Casting



Steps Involved →

① Pattern creation
- Two piece metal pattern of desired shape.

② Mould creation
- each pattern half heated to 175 - 370°C & coated with lubricant
- clamp heated pattern to dump box
- invert dump box to coat sand - resin mixture
- heated pattern form shell around the pattern.
- shell is cured in oven & halves joined

③ Mould Assembly
- shell placed in flask, supported by backing material.

④ Pouring
- molten metal is poured from ladle into mould cavity.

⑤ Cooling
- allowed to cool & solidify

⑥ Casting Removal
- mould broken & casting removed

⑦ Trimming
- remove excess material

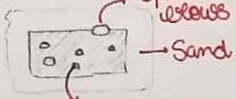
→ Advantages
- Very smooth surface
- Reduce cleaning & machining cost.
- Rapid production rate
- complex shape can be produced.

→ Disadvantages
- Resin binder → expensive
- Initial cost of pattern is high.

→ Application
- used for parts that require high precision.

Defects in Casting

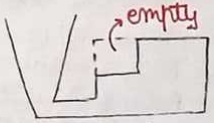
① Blow holes (large holes)
- cavity in casting
- upper surface → open blows
- concealed in casting → blow holes
- due to trapped gases in metal
- due to hard ramming, low permeability, improper venting, etc.



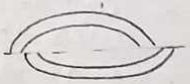
② Pin holes / Gas holes (small holes)
- due to hard ramming, low permeability, improper venting, etc.

③ Inclusions
- undesired foreign materials
- Eg → slag, oxides, dirt, etc.

④ Misrun (Incomplete Penetration)
- incomplete filling of cavity
- caused due to inadequate supply of molten metal & improper designed gates.



⑤ Mismatch
- shift of individual part of casting w.r.t. other.
- due to inexpert assembling of two halves of mould.



⑥ Flashes or Fins
- Thin projections of metal
- due to loose clamping of mould, insufficient wt. on top of mould.



⑦ Hot Tear
- High residual stress causes crack.
- due to poor casting design.

⑧ Swell
- due to high liq. metal pressure.



Core

→ Function:
- required to create holes, recesses, projection, undercuts & internal cavities.

→ Making:
- made from sand, metal & ceramic
- In core oven @ 200 - 250 °C

→ Characteristics:
- strength to withstand force of molten metal.
- permeable than mould
- have high refractoriness
- smooth for good finish
- should be capable of collapsing after being solidified.

→ Types of Cores

Ⓐ According to state core

① Green sand core
- made from ordinary moulding sand
- restricted to simple processes.

② Dry sand core
- moulding sand with controlled conditions.
- ample core irons are used for strength
- baked until perfectly dry.

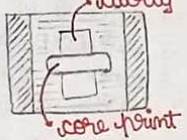
③ Oil Sand core
- chiefly of sea shore and other silica sands as binding medium, such as linseed oil, molasses and cereal flour.
- baked before handling

④ Loam Cores
- perforated iron cast or steel barrels, coated with loam.
- baked perfectly dry.

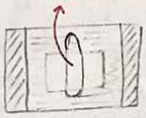
⑤ Metal Core
- made of steel
- used in non-ferrous casting
- act as densers & impart fine-finish.
- used for production of iron casting, produce white hard skin.

Ⓑ According to position of core

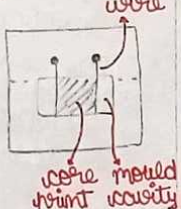
① Horizontal Core
- horizontal in mould
- supported in core seats at both end.



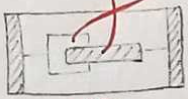
② Vertical Core
- vertical in mould
- more taper at top and at bottom.



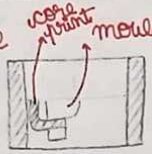
③ Hanging or cover core
- hangs & covers the mould and rest on seat made in drag.
- provided with hole, for molten metal reach cavity.



④ Balanced core
- one end supported
- used for creating blind cavity.



⑤ Drop or stop-off core
- hole, recesses or cavity → not along parting line.



Centrifugal Casting
- used for distributing the mould around its circumference.
- rotation is controlled until molten metal solidifies.
- break mould & take out casting.

→ General use :
To produce large sized pipes without using core.

→ Three Types :

Ⓐ True Centrifugal Casting
- Parts produced → pipes, tubes, bushing and rings
- axis : usually horizontal, vertical for short-workpieces.
- mould made up of steel, iron or graphite or may be coated with refractory lining to increase life.
- Casting of good quality, dimensional accuracy & external details are obtained.

Ⓑ Semicentrifugal Casting
- used to produce solid casting
- Mould are designed with risers at center to supply feed metal.
- d metal in final casting > in outer sections than at center of rotation
- used for parts where center of casting is machined away.
- Eg → wheel & pulley

Ⓒ Centrifuge Casting
- use centrifugal acc" to force metal into mould.
- axis of rotation of mould and casting are different.
- low rotational speed
- thin walls & intricate shapes

→ Advantages :
- quick & economical
- risers, feed heads, cores eliminated
- ferrous & non ferrous castings
- dense & fine grained structures
- good surface finish

→ Disadvantages :
- metal composition not uniform
- only symmetrical shapes
- limited to small intricate castings.

mechanical properties of centrifugal casting are superior than sand castings but gravity segregation is faced in some alloys.

Manufacturing Processes
Unit - 2

Welding
⊙ metal joining process which produces coherence of material by heating them to suitable temperature with or without application of pressure or use of filler metal.

→ Advantages:
⊙ Permanent joint
⊙ Low cost
⊙ flexibility in design
⊙ different metals can be joined
⊙ repair work

→ Disadvantages:
⊙ require skilled labour
⊙ residual stress remains
⊙ distortion in metal can take place.

→ Classification:

Ⓐ Fusion Welding
⊙ Gas welding
⊙ Electric Arc welding
⊙ Thermal welding

Ⓒ Miscellaneous Welding
⊙ Tungsten Inert Gas (TIG)
⊙ Metal Inert Gas (MIG)
⊙ Submerged welding
⊙ Electro-slag

Ⓑ Pressure Welding
⊙ Resistance
⊙ Butt - Resistance
⊙ Flash Resistance
⊙ Spot resistance
⊙ Projection resistance
⊙ Percussion resistance

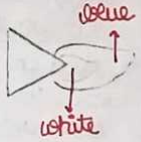
Ⓓ Modern welding
⊙ Electron beam
⊙ Ultrasonic
⊙ Laser beam
⊙ Explosive

Ⓔ Allied Processes
⊙ Soldering
⊙ Brazing

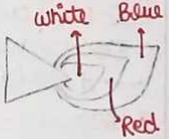
Fusion Welding
Ⓐ Gas welding
→ O_2 & C_2H_2 (Acetylene) combustion welding

→ Types of Flames:

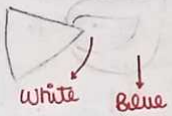
① Neutral Flame
⊙ ratio of $C_2H_2 : O_2 \rightarrow 1:1$
⊙ Temperature - 3100°C
⊙ used to weld carbon steel, cast iron, Al, etc.



② Carburising Flame
⊙ excess C_2H_2
⊙ T - 2900°C
⊙ used to weld medium carbon steel, nickel, etc.



③ Oxidising Flame
⊙ excess O_2
⊙ produces roaring sound
⊙ used to weld copper, brass, bronze, etc
⊙ T - 3300°C



→ Equipments for gas welding

Ⓐ O_2 Cylinder
⊙ made of steel & painted black
⊙ O_2 compressed at 1800 - 2000 PSI.

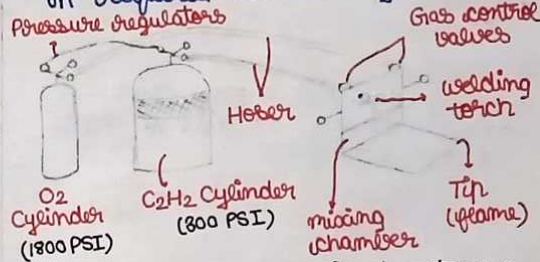
Ⓑ C_2H_2 Cylinder
⊙ maroon in colour
⊙ highly explosive → wetted with acetone
⊙ P: around 300 PSI

Ⓒ Welding Hoses
⊙ Hoses → rubber pipes use to connect cylinder.
⊙ black or green → O_2
red or maroon → C_2H_2
⊙ strong, durable, non-porous & light in weight.

Ⓓ Check valves
⊙ installed between hoses & torch to prevent backflow of gases

Ⓔ Pressure regulator
⊙ control working P of O_2 & C_2H_2

Ⓕ Welding Torch
⊙ allow O_2 & C_2H_2 to allow flow in required volume & ignite at tip.



→ Advantages:
⊙ Portable
⊙ Easily available
⊙ versatile process
⊙ adapt to diff. types of joint

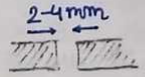
→ Disadvantages:
⊙ skilled labour required
⊙ reactive metal → not suitable
⊙ less safety

→ Application:
⊙ Joining ferrous & non-ferrous material
⊙ used in industry & sheet metal fabrication.

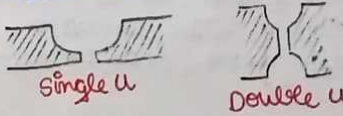
Types of Joint

① Butt Joint

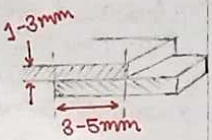
Ⓐ Square Butt Joint
⊙ thickness < 5mm
⊙ gap: 2-4 mm



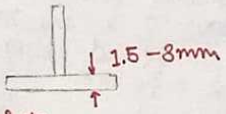
Ⓑ Single U & Double U
⊙ thickness > 20 mm



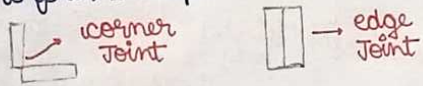
② Lap Joint
- thickness < 3mm
- plates overlap
- adapted when the width of lap : 3 to 5 times thickness



③ T – Joint
- thickness : 10 – 20 mm



④ Corner / Edge Joint
- Corner Joint → meet in corner to form L-shape
- Edge Joint → Joint at edges



Position of welding
- Vertical
- Horizontal
- Flat
- Overhead

Fusion Welding (Cont.)

Ⓑ Electric Arc Welding

→ Two types of electrodes used :

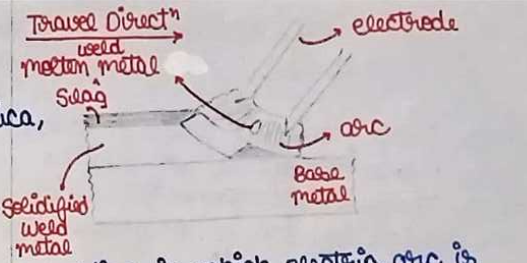
① Bare electrode
- cheaper
- poor quality products
- require high skill
- rarely used practically

② Core electrodes / Coated electrodes
- carry a core of bare metallic wire provided with coating on outer surface.
- mild steel → mostly used
- other materials used → low alloy steel, nickel steel, chromium – molybdenum steel, vanadium steel, aluminum, phosphorus – bronze, etc.

→ Electrode covering
- helps in slag formation & metal refining
- common ingredients → asbestors, mica, silica, titanium dioxide, iron oxide, magnesium carbonate, calcium carbonate & different aluminers.
- used for reducing atmosphere – cellulose, calcium, carbonate, wood flour, starch, dextrin, etc.
- Iron powder → high deposition rate
- Ferromanganese & manganese oxide ↳ provide alloying elements
- Potassium silicate & Potassium titanate ↳ principle stabilizers.
- for light & medium coated electrode, thickness of covering vary from 10 – 55% of total diameter of electrode

→ Selection of electrodes :
depends upon :
① Availability of Current – AC or DC
② Composit^{n} & Thickness of base metal
③ Welding Position
④ Expected physical properties of welded joint
⑤ Amt. of penetration required in welding

→ Principle of Arc
- suitable gas is kept betw the work and electrode.
- high current passed through the circuit
- electric energy → heat energy @ T = 3000°C to 4000°C
- heat melts edges to be weld & molten pool is formed.
- On solidificat^{n}, welding joint is obtained.



- Welding, in which electric arc is produced to give heat, for purpose of joining two surface, is called Electric Arc welding.
- Arc → flame of intense heat that is generated from electric current.

→ Equipments of Arc welding

Ⓐ Workpiece Clamp
- used to transfer current from power source to electrode holder and produces arc to the workpiece

Ⓑ Electrode Holder
- have the capacity of clipping the electrode manually to receive current & create arc

Ⓒ Clipping Hammer
- wooden or rubber handle
- used to reduce slag while welding

Ⓓ Wire Brush
- used to clean dirt & to wipe away rust from metal

Ⓔ Protective Shield
- protect eyes from harmful radiation

Ⓕ Welding Transformer
- step down transformer
- recieves current from main supply at 400 – 440 V
- transform it to required voltage for welding.

→ Advantages
- ⊙ Simple welding equipment
- ⊙ Inexpensive power source
- ⊙ Fast Process
- ⊙ used for multiple functions
- ⊙ Give superior Temp. at point of welding.

→ Disadvantages
- ⊙ require skilled operators
- ⊙ not used for thin metals
- ⊙ cannot weld reactive metals

→ Application
- ⊙ used in repairing broken part of machines
- ⊙ used in ship building.

→ Working:
- ⊙ Intensive heat at arc melts tip of electrode
- ⊙ Tiny drops of metal enters arc stream and deposited on parent metal
- ⊙ after this, slag is formed (slag serves as an insulation against air contamination)
- ⊙ Cooling, the oxide layer is removed by chipping hammer & cleaned with wire brush before next pass.

→ Use of AC in EAW
- ⊙ Cheaper
- ⊙ Easier maintenance
- ⊙ No moving parts
- ⊙ Bare electrode cannot be used
- ⊙ Only work in AC
- ⊙ Not preferred for sheet metal welding.

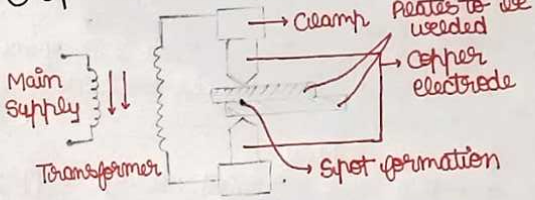
→ Use of DC in EAW
- ⊙ more moving parts
- ⊙ better suitable at low current
- ⊙ Easier maintenance of short arc
- ⊙ Can work in both
- ⊙ Different fixed polarities can be used for weld.
- ⊙ easier starting of arc & steady rate

Pressure Welding

Ⓐ Resistance Welding
- ⊙ weld is made by combinat" of heat, pressure & time.
- ⊙ It is defined as resistance of material to be welded, to current flow that causes localization of heat to the part that has to be welded.

$$H = I^2RT$$

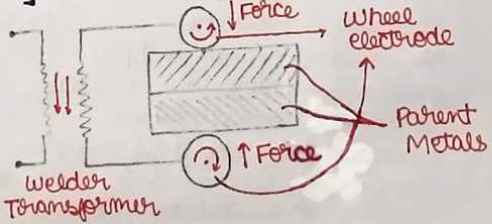
Ⓑ Spot Resistance Welding





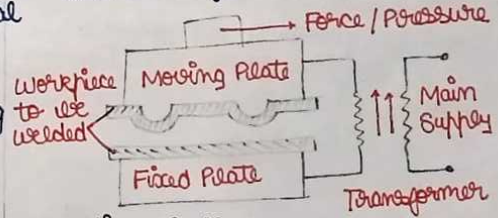
- ⊙ Principle same as resistance welding
- ⊙ used to weld various metal sheet of thickness from 0.5 - 3mm.
- ⊙ Process uses two copper shaped electrode to concentrate welding current and force other materials to be welded. This result in spot formation or nugget of welded material after current is removed.

Ⓒ Resistance Seam welding
- ⊙ a series of overlapping spot welding made progressively along the joint by two rotating wheel electrodes.
- ⊙ Force is applied by the wheel to produce continous joint.
- ⊙ Principle same as resistance welding



Ⓓ Percussion Welding
- ⊙ consist of holding part at small distance with their end faces opposite to each other, bringing them closer at fast speed after switching on current, thus creating arc betw their end faces just before they come in contact with each and complete the weld under impact.
- ⊙ some of the metal may squeeze out of the joint, but is very small.

Ⓔ Projection Resistance Welding
- ⊙ small projections are raised on one side of plate with punch & die assembly & other plate is placed.
- ⊙ During welding, projection collapse owing to heat & pressure, and part to be joined are brought in close contact.
- ⊙ projection act to localise the heat & weld is formed



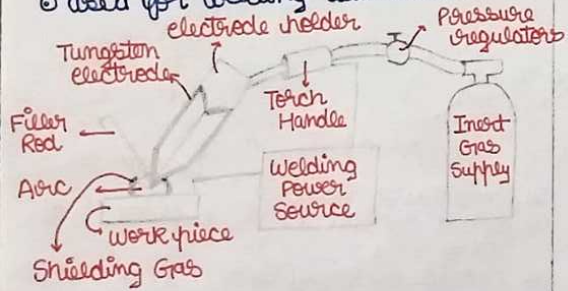
Function of flux
- ⊙ shield around the arc & protect from atmosphere
- ⊙ provide alloying element to molten metal
- ⊙ helps in stabilising the arc
- ⊙ control the bead shape

Tungsten Inert Gas (TIG) Welding

- Uses non-consumable tungsten electrode, mounted in electrode gun.
- Electrode welder is designed to furnish the flow of inert gas around the electrode and around the arc.
- Tungsten electrode is alloyed with thorium or zircona, used to stabalise the arc.
- If filler rod required, a welding rod is fed into welding zone & melted with base metal.
- Welding is done by striking the arc btw work piece & tungsten electrode
- Both AC & DC power can be used:

(a) DCSP (Direct Current Straight Polarity)
→ Tungsten electrode (-ve), workpiece (+ve)
- used for mild steel, stainless steel & titanium (Mg).
- used for more thicker material as it has more heat. (workpiece)

(b) DCRP (Direct Current Reverse Polarity)
→ Tungsten electrode (+ve), workpiece (-ve)
- used for welding aluminium.



Metal Inert Gas (MIG) Welding

- Semi-automatic or automatic arc
- continous and consumable wired electrode & a shielding gas are fed through welding gun.

→ Equipments of MIGW →

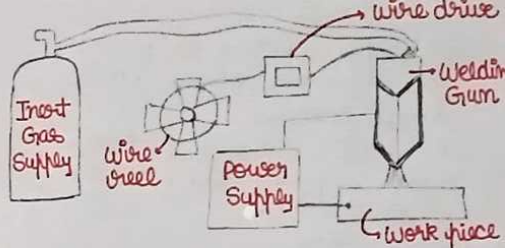
Ⓐ Welding Gun & Wired feed Unit
- consist of control switch, contact tip, power cable, gas nozzle, electrode & gas horse
- control switch → pressed, initiate wire feed resulting formation of electric arc btw electrode & workpiece
- wire feed unit supplies electrode to work piece

Ⓑ Power Supply
- mostly uses constant voltage supply
- Generally DC is used.

Ⓒ Electrode
- continous spool of wire electrode reel is used, which is consumable

Ⓓ Shielding Gas
- protect weld from outer atmosphere
- generally used are → Ar, He, etc.

→ Working
- consumable electrode in the form of wire reel, feed at const. rate, through feed rollers.
- welding torch is attached to gas cylinder to provide necessary inert gas.
- electrode & workpiece are connected to welding power supply
- power supplies are generally constant voltage type.
- current from welding machine is changed by changing the rate of feeding of electrode wire.



Common Points btw TIG & MIG

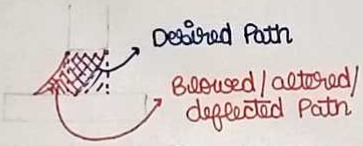
- Shielding Gas
- Gas flow & its control
- cleaning the weld is imp.
- No flux required.

Difference btw TIG & MIG

TIG	MIG
⊙ non consumable tungsten wire	⊙ consumable cont. coil electrod
⊙ produce only arc	⊙ produces arc as well as filler metal
⊙ Slower than MIG	⊙ Faster
⊙ skilled operator	⊙ No need
⊙ filler metal handled manually	⊙ all automatic
⊙ DCSP & DCRP used	⊙ DCRP is used
⊙ water cooled	⊙ No need
⊙ Tungsten may cause contaminat" when in contact with molten metal.	⊙ No such need
⊙ Penetrat" not so deep.	⊙ Deeper penetrat"s

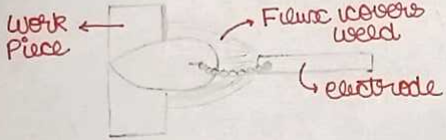
Arc Blow

- deflection of arc from its intended path.
- due to magnetic forces acting on arc as current flows through electrode.
- greater at begining / end of weld
- controlled by changing current and arc length.



Submerged Arc Welding (SAW)

- between base metal and arc
- Pressure is not applied
- filler metal is obtained from electrode and at same time from supplementry welding rod
- Shielding is obtained from blanket of granular flux, which is laid directly over weld area. The weld is submerged under this layer of flux, hence known as SAW.
- normally operated in automatic & mechanised mode
- Current range : 200 – 1500 A



→ Advantages
- High deposition rate
- Deep weld penetration
- Weld bead are smooth

→ Limitations
- Since weld can't be seam, more difficult to guide electrode
- mainly done in flat position
- Not suitable for metal less than 8mm thickness.

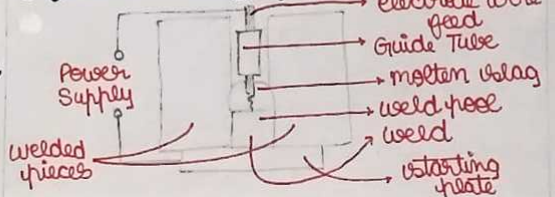
Electro-slag Welding

→ Principle :

Arc + Resistance ; DCRP is used

- Arc created b/w starting block and copper coated electrode wire powered by addition of flux.
- welding progresses in vertical direct" and due to that temp. rises on increasing.
- a pair of water cooled shoes is being provided on the sides to avoid any spillage of liq. metal to the sides.
- once the Temp. of molten pool reaches a particular value, arc is switched off and molten pool is so hot that it starts consuming the wire without any arc.
- Temp is controlled, to minimise the heat affected zone.
- If heat affected zone increases, it may cause distortion of plate.
- The point when molten metal is not able to consume any more wire, arc is restarted again.

→ Application :
- used to join two thick plates along edges
- building the ship
- can be used to weld plates of thickness 20 – 40mm
- used in heavy ferrous industry



Mode of metal transfer in MIGW

	Spray Transfer	Globular Transfer	Short Circuit Transfer	Pulse Spray Transfer
Droplet Size	Smaller than wire diameter	3x wire diameter	direct transfer by contact	much smaller than wire diameter
Current	220A+	150–400A	180A–	220A+
Force used	Electro-magnetic Force	Use gravity to transfer metal	Lower voltage compared to spray arc.	Electro-magnetic force dominate

Welding Defects

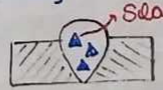
1. Porosity

→ Cause →
- Gas released during melting of weld area trapped during solidification.
- Chemical reactions
- Due to contaminants



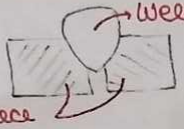
→ Reduced by →
- proper selection of electrode & filler metal
- preheating weld area
- proper cleaning
- reducing welding speed

2. Slag Inclusion
- oxides, electrode coating material, dirt and dust trapped in weld zone



→ Prevention →
- proper cleaning of weld surface
- providing enough shielding gas

3. Incomplete fusion or penetration
- occurs when depth of welded joint is insufficient.



→ Prevention:
- ⊙ increasing heat input
- ⊙ reducing travel speed
- ⊙ ensuring the surface to be joint properly.

4. Under-Cutting
- ⊙ result from melting away waste metal and generat^n of grooves in shape of sharp recess or notch



5. Overlap
- ⊙ surface discontinuity
- ⊙ due to →
  - ① Poor welding process
  - ② improper select^n of material



6. Cracks
- ⊙ due to →
  - ① Thermal stress in weld zone
  - ② Inability to contract during cooling



7. Spatter
- ⊙ small droplets of molten metal and get solidify in metal surface.



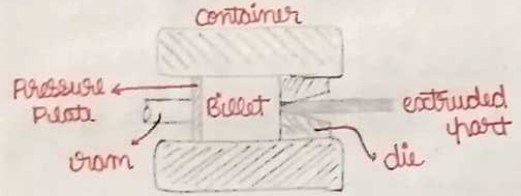
Unit-3
Manufacturing Processes

Extrusion
- Cross section of metal is reduced by forcing the metal through a die orifice under high pressure
- Application → used to produce cylindrical bar, tubes & sections of any regular or irregular types.
- Classification →

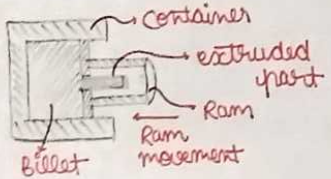
① Hot Extrusion
→ Forward or direct extrusion
→ Backward or indirect extrusion

② Cold Extrusion
→ Impact extrusion
→ Hydrostatic extrusion

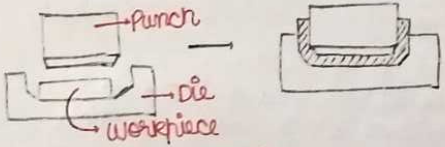
Hot Extrusion
Ⓐ Forward or Direct extrusion
- metal billet is placed in a container and extruded through the die with help of ram.
- Ram & workpiece move in same direction.
- pressure pads in contact with billet and ram
- relative motion betw container wall creates high friction. Hence, power is relatively high.



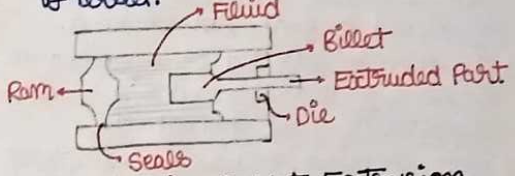
Ⓑ Backward or Indirect Extrusion
- hollow ram compresses metal through a die in a direction opposite to ram motion.
- No relative motion, lower friction, power required is low.
- Limitation: due to hollow ram, limited load can be applied & only small section can be extruded



Cold Extrusion
Ⓐ Impact Extrusion
- used for soft ductile materials operations
- To form a component, workpiece is kept on die and punch strike the workpiece against the die.
- Due to pressure, metal extruded to the gap between punch and die opposite to punch movement & slide along side wall
- Thin walled cans, collapsable tubes are formed.



Ⓑ Hydrostatic Extrusion
- High pressure fluid is used to apply the force to extrude the work piece through a die.
- Friction not present
- No direct force betw Ram & billet
- Fluid transfer force from Ram to billet.



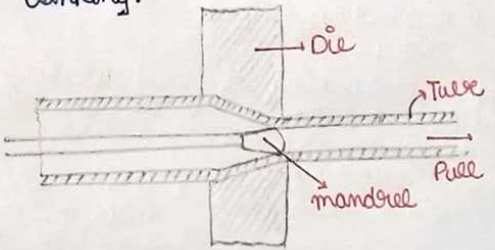
Cold Extrusion > Hot Extrusion
- Improved mechanical properties
- Good control of dimensional tolerances
- Improved surface finish
- No billet heat required
- Production rates & costs are competitive

Hot working v/s Cold working

Hot working	Cold working
carried at T > recrystallizat^{n} T (but below MP)	T < recrystal^{n} T
refines metal grains	doesn't refine metal grains
Hardness of metal ✗	hardness in workpiece
No internal, residual stress	Internal & residual stress ✓
Stress req. to cause deformat^{n} is less due to high deformation temp.	Stress req. to cause deformat^{n} is high due to low deformat^{n} temp.
increases impact strength & elongation	reduces impact strength & elongat^{n}
Poor surface finish	Good surface finish
maintain closed dimension tolerance	High dimension tolerance
existing cracks & blow holes eliminated	Not possible to eliminate

Tube Drawing

- cold working process, used to draw tubes or pipes by using hollow seamless and welded tube shells produced through hot working.
- operation is started by swaging the diameter of starting end to a smaller diameter to pass through die.
- after passing, end is gripped in tongs and connected to draw bench for tube drawing.
- outer diameter → controlled by die
  internal diameter → by internal mandrel
- If mandrel absent, called as tube sinking.



Forging Process

→ Steps to be followed for forging

Step 1 → Heat metal workpiece, generally in furnace

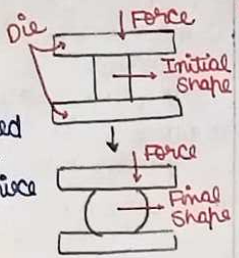
Step 2 → Keep hot metal on anvil

Step 3 → apply hammer force on hot metal. Desired shape ✓

→ Definition: In forging operation, force is applied on heated metal manually or by means of power hammer. This applied force squeeze the metal into two or more dies & change size & shape of workpiece.

Types of Forging

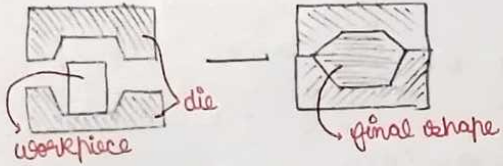
① Open die Casting



- impact is delivered by hammer
- operator obtains desired shape by orienting & positioning the workpiece into the recess.

② Closed die Casting

- workpiece completely fill the die cavity without any loss of material.
- accurate workpiece sizing is required to obtain forging



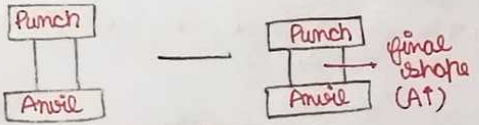
Forging Operations

① Drawing out

- cross sectional area reduces & length of stock increases

② Upsetting Out

- cross section increases & length of stock reduces



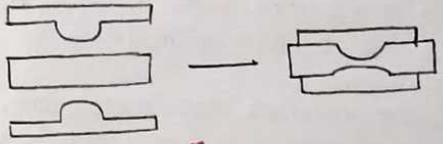
③ Heading operation

- done at one end of stock to produce large cross section
- Heads of bolts, nails & screw are formed.



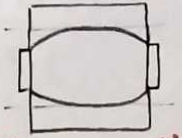
④ Fullering Operation

- displacing metal away from centre to reduce thickness in middle & increases the length.



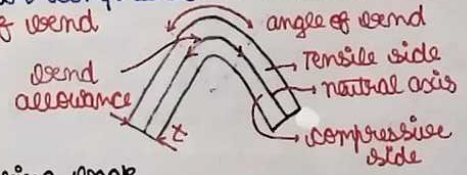
⑤ Edging Operation

- material is displaced from edges to inside i.e. into two cup shaped die to create bulge



⑥ Bending Operation

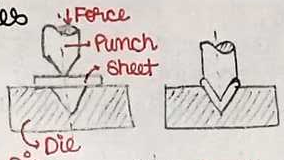
- directional change produced in workpiece
- Bending → plastic deformation of metal about a linear axis with little or no change in surface area.
- location that is neither stretched nor compressed → neutral axis of bend



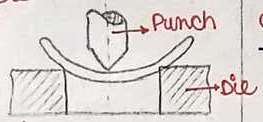
- Spring back
  → Plastic deformation followed by some elastic recovery when load is removed. This recovery is spring back.
  → bend angle ↓
  → compensated by over bending

⊙ Bending Dies

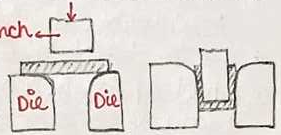
1. V-Die
⊙ used in angle bending



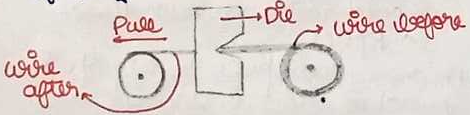
2. Arc Bending Die
⊙ produces geometry by simple three point bending.



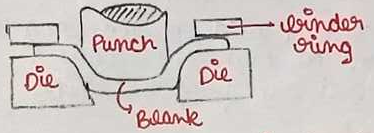
3. U-Die
⊙ U-die is used to bend sheet in U shaped.



Wire Drawing
⊙ rod passed through conical die to obtain wire
⊙ the front end of rod is pointed by cutting or squeezing, freely enter through die opening.
⊙ it is pulled to sufficient length to attach to operated reel.
⊙ reel is rotated at desired rate to draw the wire.
⊙ contact region of die are made of tungsten carbide or diamond



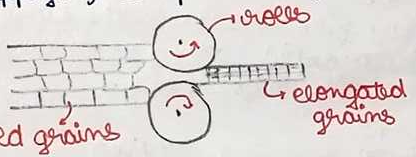
Deep Drawing
⊙ used to form containers or object with depth from flat sheet.
⊙ accomplished by use of hydraulic or mechanical press.
⊙ require double action force
⊙ one force → hold sheet
other force → by punch



⊙ Advantages
→ product" of shallow or deep parts
→ High product" rate

⊙ Disadvantages
→ High cost of tools used in deep drawing

Rolling of metals
⊙ reduce thickness or change cross section of long work piece by applying compressive forces



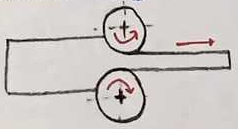
→ Principle
⊙ metal emerges from the roll at higher speed than it enters.

$Q_i = Q_f$ (quantity is constant)

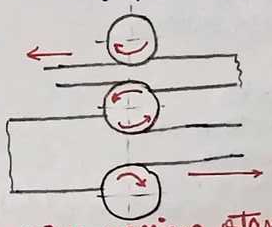
$A_1 v_1 = A_2 v_2 \rightarrow \frac{A_1}{A_2} = \frac{v_2}{v_1}$

→ Arrangement of rolls in rolling mills are called rolling stand.
Most common arrangements are →

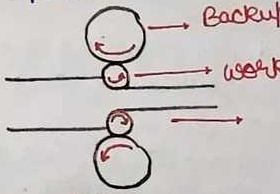
Ⓐ Two High Non-reversing rolling stand
⊙ two rolls positioned one over the other, supported by bearing
⊙ unidirectional
⊙ distance may be adjusted betw them.



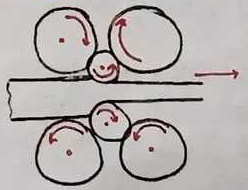
Ⓑ Three High rolling stand
⊙ eliminate need of reversing rolls
⊙ workpiece is fed in betw upper & middle rolls in direct" opposite to feeding of same in lower & middle roll.
⊙ High rate of production



Ⓒ Four High rolling stand
⊙ back up rolls provides necessary rigidity to smaller rolls, prevent deflection
⊙ used for hot & cold rolling of wide plates & sheets



Ⓓ Cluster Mill
⊙ two smaller diameter rolls & four or more backup rolls.
⊙ Foil is always rolled on cluster mills.



Forging Defects

→ Causes →
1. Improper heating
2. Incorrect die design
3. uneven cooling of stock
4. defects in original metal
5. improper forging operations and conditions

① Cold Shuts
- ⊙ short cracks
- ⊙ caused due to sliding metal surface against itself.
- ⊙ avoided by rounding sharp corners.

② Pitting
- ⊙ caused by incomplete scales, not removed from die cavities and worked as surface forging
- ⊙ produces shallow depressions on the surface.
- ⊙ avoided by cleaning of dies and proper control of forging Temp.

③ Incomplete filling of Dies
- ⊙ due to incomplete filling of closed impression die cavity.

④ Die Shift
- ⊙ caused due to upper & lower die misalignment
- ⊙ due to loose wedges
- ⊙ avoided by fixing wedges rigidly

⑤ Fins & Rags
- ⊙ small projection or pieces of loose metal driven into surface.
- ⊙ avoided by proper die design.

⑥ Burnt & Overheated metal
- ⊙ metal is heated for too long time or at too high temperature
- ⊙ avoided by proper heating

⑦ Cracks
- ⊙ Both transverse & longitudinal cracks
- ⊙ due to improper heating of forgings, bad quality of alloy & improper cooling of alloy steel.

⑧ Flakes
- ⊙ internal breaks and ruptures caused due to rapid cooling
- ⊙ avoided by proper cooling.

⑨ Hairline cracks
- ⊙ very fine cracks on the surface
- ⊙ due to rapid cooling of forging & defective metal piece.

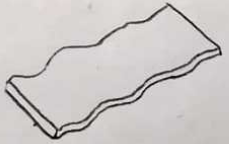
⑩ Dents
- ⊙ due to careless working

Rolling Defects

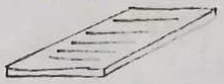
① Surface Defects
- ⊙ scale, cracks, scratches
- ⊙ due to:
  - → impurities & inclusion in billet
  - → improper condition related to material preparat^{n} & rolling operat^{n}

② Internal Structural Defects

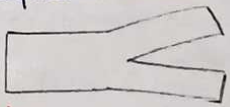
Ⓐ Wavy edges
- ⊙ caused by roll bending
- ⊙ strips are compressed more than central portion resulting in more elongat^{n} at edges than center



Ⓑ Zipper Cracks
- ⊙ occur at center of sheet due to roll bending and poor ductility of material at rolling T.



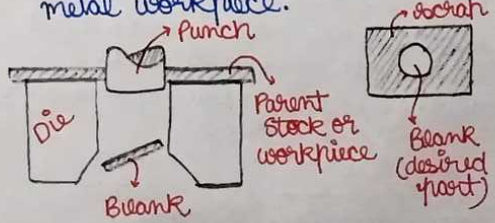
Ⓒ Alligatoring
- ⊙ encountered due to non uniform distribution during rolling or due to defect in billet.
- ⊙ sheet splits along a horizontal plane & follows the path of their respective rolls.



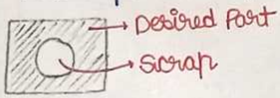
Ⓓ Folds
- ⊙ occur during plate rolling due to very small reduct^{n} per pass.

Sheet metal Operations

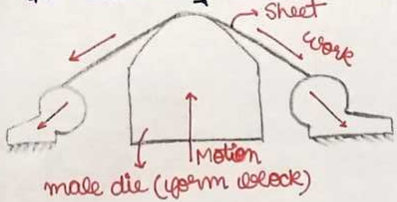
① Blanking
- → metal working process during which a metal workpiece is removed from metal sheet, then it is punched.
- → removed metal is new metal workpiece
- → primary workpiece is scrap & punched part is desired new metal workpiece.



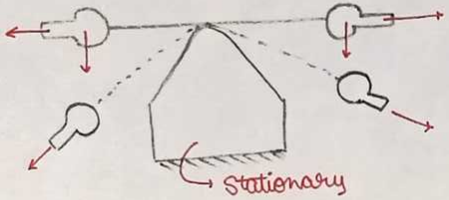
② Punching
→ punched out part is scrap and primary sheet left is useful part.
→ used for cutting holes, slots in sheet metal plates or parts.



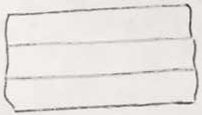
Stretch Forming
⊙ process of production of contour parts by stretching metal sheets over a shaped form block.
⊙ Stretch forming strains the metal beyond elastic limit, to give workpiece a permanent set.
→ Two Types :-
① Form Block Method
⊙ grips are stationary and form blocks moves upwards to provide the necessary tension and motion.



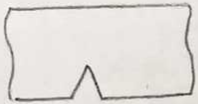
② Movable Jaw Method
⊙ first stretch the workpiece and then wrap it around the form block
⊙ In this machine, jaws can be easily and rapidly positioned in close proximity to form block edges to eliminate excessive material waste.
⊙ It has advantage over form block method as spring back is either greatly reduced or completely eliminated, since direct bending stress are never introduced.



Slitting
⊙ used to cut wide coiled sheet into no. of narrow strips through slitter.



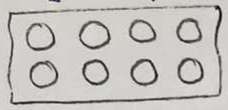
Notching
⊙ a specified portion of metal toward end of stock is removed.



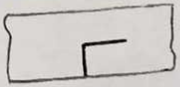
Trimming
⊙ small amount of unwanted material (known as Flash) is removed.
⊙ Flash is generally formed during die casting & drop forging

Shaving
⊙ small amount of metal is removed from edge of already blanked part

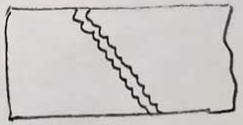
Perforating
⊙ large no. of closely spaced holes are identical & arranged in regular pattern.



Lancing
⊙ cutting operation that forms a line cut in the metal



Nibbling
⊙ used to remove the metal in small increment.
⊙ In this operation, contour is cut out from sheet metal by producing a series of overlapping slit or notches either from outer boundary or from punched holes.



Metal Spinning
(for symmetrical round or cylindrical object)
⊙ transform a flat circular blank or disk shaped workpiece into axially symmetrical round shapes.
⊙ accomplished by application of lateral force by a roller positioned against the surface of the disk blank and a mandrel, spinning on its axis at high speed

⊙ mandrel serves as support for the workpiece and help in giving its final shape.

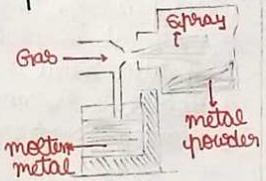
Unit - 4
Manufacturing Processes

Powder Metallurgy
- deals with the production of metal and non-metal powders and subsequently manufacture of components by these powders.

Powder Metallurgy Processes
→ Primary Operations

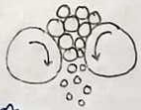
1. Producing metal powders

A. Atomization
- process of metal spraying against a stream of compressed air or inert gas.
- used for metals with low temp. metals such as Pb, Al, Zn & Tin.



B. Reduction
- compounds of metals usually oxides like iron oxides are reduced with CO/H at temp below MP.

C. Crushing
- passing the metal parts against two rollers to form powder.



D. Milling
- balls hit the material and break it in powder form.

E. Shotting
- pouring molten metal through a sieve or orifice.
- spherical or pear shaped powder particles

F. Electrolysis
- desired metal at anode of electrolytic cell
- anode slowly dissolves and gets dissolved and deposited at cathode, from where deposit is removed, washed and dried.

2. Mixing / Blending of metal powders

A. Blending : mixing powder of same chemical composition but different sizes.

B. Mixing : combining powders of different chemistries

3. Compacting (Pressing of metal powders)
- Pressing the powders into desired part, shape as closely as possible to final dimensions.
- Pressure is very high
- Compacting processes are:
  1. Die pressing
  2. Roll pressing
  3. Extrusion

4. Sintering
- It consist of heating pressed metal compacts in batch or continous furnace to a temp. below MP of material.
- increases strength & hardness
- T is 70-80% of melting Temp.

5. Finishing
- final operation with which improves several dimensional tolerances, physical properties & smooth finish

→ Secondary / Finishing Operations

1. Sizing
- used pressing the sintered part to improve dimensional accuracy.

2. Coining
- used pressing to press details into its surface.

3. Impregnation
- oil fills the pores of the part

4. Infiltration
- pores are filled with a molten metal

Application
1. Metal filters
2. Cutting Tool & dies → Turbine blades
3. Machinery Parts → Carbon Boron Nitride Tool
4. Bearing and Bushes → Self lubricating Bearing
5. Magnets

Advantages
- High dimensional accuracy achieved
- No wastage
- Porous parts can be produced
- eliminates numerous matching operations
- economical for mass production

Limitations
- High tooling cost
- difficult to manufacture long parts
- not economical for small scale production
- expensive raw materials
- poor ductility of products

Plastic
- polymers of long carbon chains

→ Types of Plastic :

Thermoplastic	Thermosetting Plastic
- can be deformed easily on heating	- can't be softened by heating
- synthesised by addition polymerization.	- synthesized by condensation polymerization.

- low MP and Tensile strength
- PVC, Polythene
- manufacturing toys, lenses, containers

- high MP and Tensile strength
- Bakelite & melamine
- floor tiles, fabrics, kitchen-ware

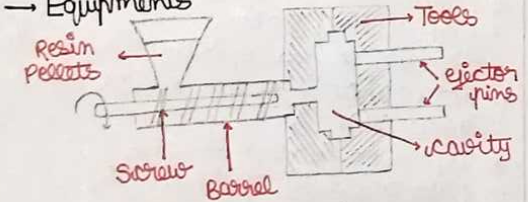
→ Advantages

1. Thermoplastic
   - flexible product design
   - corrosion resistant
   - water resistant
2. Thermosetting Finish
   - High impact resistant
   - Good electrical insulation

Injection Moulding

- produce parts are components by injecting molten materials into mould cavity.
- for materials like glass, plastic, etc.

→ Equipments



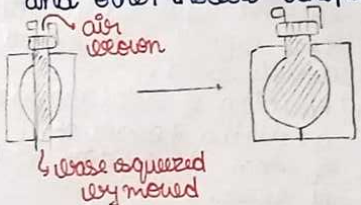
→ Advantages
- efficient high production
- low cost per part
- Low wastage
- Easily automated
- Large material choice

→ Disadvantages
- High initial cost
- Design limitation

→ Application
Toys, wire reels, packaging, containers

Blow Moulding

- forming hollow plastic parts
- used for forming glass bottles and other hollow shapes

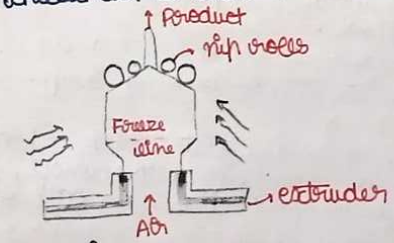


Rotating moulding

- hollow mould is filled with powdered plastic resin.
- mould is rotated biaxially and transferred into an oven.
- mould rotates and resin melts & coats the wall

Film Blowing

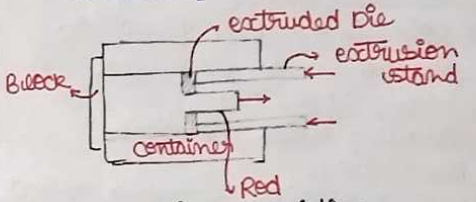
- after exiting the die, film tube which is still in plastic state, is pulled upwards by nip rollers.
- at same time, compressed air is blown into the film tube to achieve desired film thickness & width
- chilled air is blown outside the tube



Extrusion

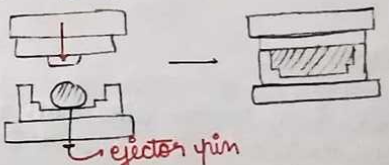
- material undergoes plastic deformation by application of force carrying that material to flow through an orifice.

→ Advantages :
- low initial cost
- continous production
- high uniaxial strength
- material thickness controlled accurately.



Thermosetting moulding

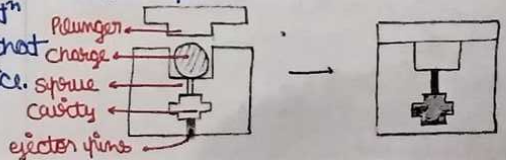
1. Compression moulding
   - material is placed directly into a heated metal
   - excess is removed post moulding
   - mostly for thermosetting



2. Transfer Moulding
   - Pressure is applied through piston
   - Process ranges low compression moulding & injection moulding

→ Advantages
- Lower maintenance cost than inject°
- durable & dimentionally stable parts

→ Disadvantages
- wastage of material
- Air can be trapped in the mould



Bonding of Thermoplastic

1. Solvent Bonding

- Used for joining injection moulded component of amorphous thermoplastic.
- Solvent dissolves the surface of two components & allows the material to flow together.
- After solvent evaporates, leaves pure m-to-m bond.

- Advantage:
  - → good distribution of stress
  - → bond could be hermic
- Disadvantage:
  - → sufficient area requirement
  - → solvent vapours may be hazardous.

2. Adhesive Bonding

- most convenient method of assembly
- possibility of joining similar or dissimilar materials
- distribute stress
- can provide hermic seal
- easier application
- five major families of adhesives
  - → classification on curetime, cure temp., resistance, application area.